Executive Summary

Employee Compensation & Salary Equity Analysis

Excel + Power Query | Portfolio Project

A descriptive analysis of compensation patterns across gender, department, job role, tenure and education.

Project Overview

Objective, data and analytical approach

5,000
Original records

4,990
Cleaned records

15
Variables

Excel + PQ
Primary tools

- Objective: identify compensation patterns and areas for deeper salary-equity review.
- Data preparation: audit and clean the dataset using Excel Power Query.
- Analysis: compare average and median compensation across key workforce segments.
- Deliverable: Excel dashboard with KPIs, charts, findings and methodology.

Key Compensation Findings

What the cleaned dataset shows

€83,201
Average base salary

€81,900
Median base salary

€90,833
Average total compensation

€89,100
Median total compensation

- Departments: average base salary ranges from €79,217 in Marketing to €87,796 in IT.
- Job roles: average base salary ranges from €63,695 for Sales Representatives to €103,334 for IT Managers.
- Tenure: average base salary increases from €76,758 for 0–2 years to €88,903 for 12–14 years.
- Education: average base salary increases from €78,375 for High School to €88,654 for PhD.

Gender Compensation Analysis

Descriptive comparison — not an adjusted pay-gap analysis

€83,291
Male average base salary

€83,112
Female average base salary

€180
Observed average difference

- Median difference: €200, with male median base salary at €82,100 and female median at €81,900.
- Interpretation: the observed difference is descriptive and does not establish a cause for the difference.
- Limitation: the analysis does not control for job role, tenure, education, performance, employment type or other relevant factors.
- Next step: perform adjusted compensation analysis to assess whether differences remain after accounting for workforce characteristics.

Data Quality & Methodology

Cleaning and validation were central to the analysis

- Duplicates: 10 duplicate Employee_ID records removed, leaving 4,990 unique records.
- Missing values: resolved where reliable source information was available; unresolved fields were retained as null where appropriate.
- Compensation validation: Total Compensation recalculated as Base Salary + Bonus and salary-band inconsistencies reviewed.
- Outliers: salary outliers identified using the IQR method and retained where there was insufficient evidence they were data errors.
- Dates: mixed Hire Date formats standardized to a consistent Date type.
- Final validation: 4,990 records with no Power Query errors.

Recommendations

How the analysis can support further HR action

- 1. Adjusted compensation analysis: control for role, tenure, education, performance and employment type.
- 2. Role and department review: compare salary ranges, internal benchmarks and salary-band positioning.
- 3. Compensation progression: examine salary progression, promotions and tenure-related increases for consistency.
- 4. Education differences: distinguish education effects from role, experience and workforce composition.
- 5. Data governance: standardize data-entry rules and introduce recurring validation checks.
- 6. Regular monitoring: refresh the dashboard periodically to identify changes and areas for investigation.

Business Value & Next Steps

From descriptive reporting to deeper workforce analytics

- Business value: transforms a messy compensation dataset into a validated, decision-ready view of workforce compensation patterns.
- Analytics value: demonstrates data auditing, Power Query transformation, formula-based analysis, visualization and dashboard design.
- Next step 1: build an adjusted compensation model controlling for relevant employee characteristics.
- Next step 2: add salary-band positioning and promotion history where available.
- Next step 3: connect the dashboard to PivotTables/PivotCharts and slicers for interactive exploration.
- Portfolio takeaway: demonstrates an end-to-end workflow from data quality through insight and actionable recommendations.